Conservation of Energy

Synopsis: Energy is one of four quantities in nature that are conserved. (The other three are mass, momentum, and electric charge.) When we say energy is conserved, we mean that energy can neither be created nor destroyed. You will learn in the following topic that momentum is one such quantity. Recall that this conservation law requires the total amount to remain constant. As a result, an object can have a varying amount of kinetic (KE) or potential energy (PE) depending on the object's position (which determines PE) and velocity (which determines KE). Still, the object's total energy or the sum of KE and PE must remain the same. For this to happen, all changes must add up to zero (an energy gain must match an energy loss), i.e., it's a "zero-sum game." For example, to conserve total energy, when an object gains PE (KE), it loses KE (PE) by the exact amount or vice versa, so the total remains the same. So here, the energy changes abide by the same zero-sum rule.

You might be wondering why I only mention KE and PE. This is because, fundamentally, ALL types of energy we know belong to PE (position-dependent) or KE (velocity-dependent). This might sound strange to you, but it's true. For example, electrical, nuclear, and chemical energies can be considered as a form of PE.

We will only describe the energies of objects that we can physically handle or easily observe, such as everyday objects, planets, or stars, rather than atoms or molecules, which are very small. An individual object's kinetic energy is $KE = \frac{1}{2}mv^2$ where KE is in J, m is mass in kg, v is the velocity in m/s, and potential energy is $PE = mgh$ where PE is in J, m is mass in kg, and h is the height from a reference point. (There are two types of PE: Gravitational (GPE) and elastic potential energy (EPE). GPE = mgh, the equation you just saw. EPE will be covered later.) Note that energy converts back and forth between KE and PE depending on the object's motion, but the total energy is always the same. However, this is only true if the object experiences no friction! When there is a friction force, the object's total mechanical energy (KE + PE, excluding thermal energy) decreases and eventually completely dissipates! When this happens, the object has no energy for motion, so it comes to rest. Total mechanical energy (KE + PE) converts to thermal (internal energy). Mechanical energy can perform useful work, such as lifting and pushing objects, but thermal energy is only heat, which can't be easily converted to push or lift objects. Thermal energy is often considered "wasted energy."

In summary, the total energy remains constant regardless of the object's motion. If there is no friction, the total mechanical energy (PE + KE) remains constant and is always maintained. If there is friction, mechanical energy converts to thermal energy, resulting in dissipation ("disappearance") or is wasted, no longer able to do useful work through pushing, pulling, or lifting an object.

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Performance Expectations: Describe quantitatively the conservation of energy. Describe mechanical energy as the sum of PE and KE. Describe the role of friction: Without friction, total mechanical energy always remains the same. With friction, total mechanical energy dissipates as thermal energy.

For PHY 151 — with calculus: Derive the conservation law by integrating Newton's second law once, and show why friction can only lower mechanical energy.

Instructions and Questions

1) The following equations describe conservation of energy. We will consider two cases.
   i) The object experiences no friction. Total mechanical energy (PE + KE) always stays the same.
   ii) Object experiences friction. Total mechanical energy gradually decreases, turning into "useless" thermal energy. Eventually, all mechanical energy can be converted into thermal energy, and the object comes to a stop.

No friction	With friction

2) Skate ramp revisited. Let's take a look at PE, KE, and thermal energies of the skateboarder:

No friction	With friction

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3) Roller coaster: Make a rough sketch of the roller coaster model. Answer the following questions:

i) Does the marble reach the other end of the roller coaster?

ii) Does the marble return to its starting position? If not, why do you think it doesn't?

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iii) How would you cause the marble to return to its starting position?

Claim:

Evidence:

Reasoning:

4) Answer the following questions on conservation of energy:

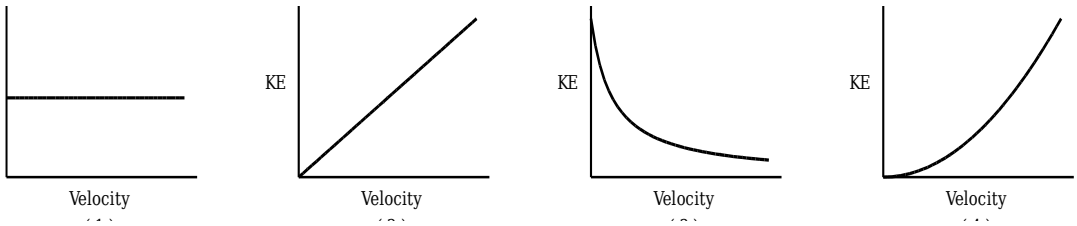
i) A 1-kilogram rock is dropped from a cliff 90 meters high. After falling 20 meters, the kinetic energy of the rock is approximately

(1) 20 J (2) 200 J (3) 700 J (4) 900 J

ii) If the speed of a car is doubled, the kinetic energy of the car is

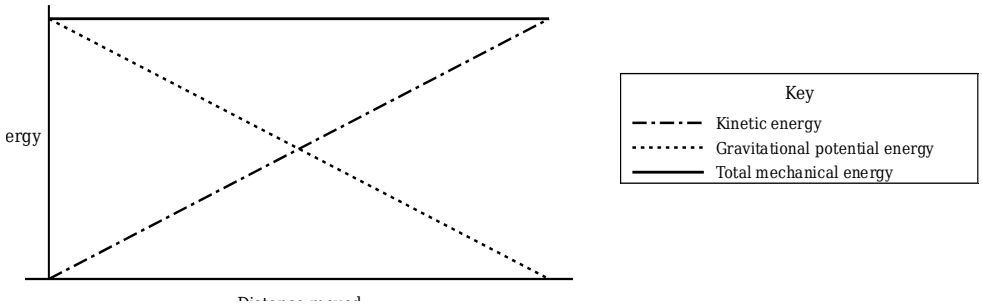
(1) quadrupled (2) quartered (3) doubled (4) halved

iii) Which graph best represents the relationship between the kinetic energy, KE, and the velocity of an object accelerating in a straight line?



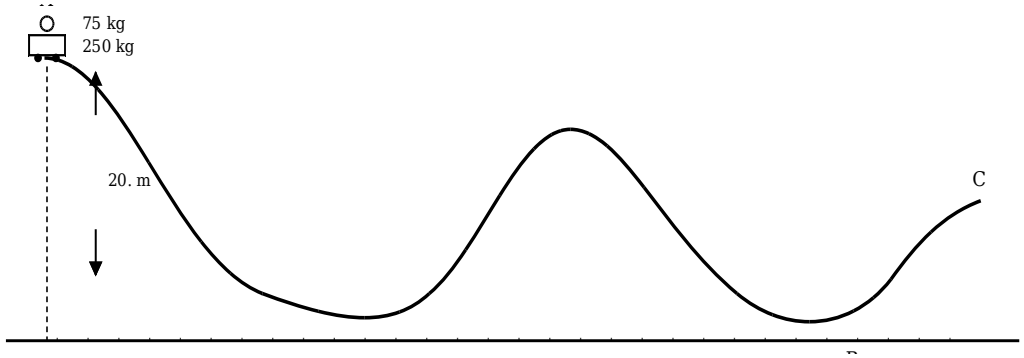
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iv) The graph below represents the kinetic energy, gravitational potential energy, and total mechanical energy of a moving block. Which best describes the motion of the block?



(1) accelerating on a flat horizontal surface (2) sliding up a frictionless incline
(3) falling freely (4) being lifted at constant velocity

v) A 250-kilogram car is initially at rest at point A on a roller coaster track. The car carries a 75-kilogram passenger and is 20 meters above the ground at point A. [Neglect friction.]



i) Calculate the total gravitational potential energy, relative to the ground, of the car and the passenger at point A. [Show all work, including the equation and substitution with units.]

ii) Calculate the speed of the car and passenger at point B. [Show all work, including the equation and substitution with units.]

iii) Compare the total mechanical energy of the car and passenger at points A, B, and C.

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Where does the law come from? Integrate Newton's second law once.
Start from m·dv/dt = F(x) for motion along a line and multiply both sides by v = dx/dt. The left side becomes d/dt(½mv²) and, when the force comes from a potential energy, F = −dU/dx, the right side becomes −dU/dt. Moving everything to one side: d/dt(½mv² + U) = 0. The sum KE + PE has zero rate of change — it is constant. That is the conservation law, and it was never an extra assumption: it is Newton's second law integrated once. Note 2 ran this in reverse (differentiate the law, get Newton). Now add a friction force of size f that always opposes the motion. The extra term on the right is −f·|v|, which is never positive, so d/dt(KE + PE) ≤ 0: mechanical energy can only go down, at a rate f|v|, and the total it loses over a trip is exactly the W_{NC} of item 1.

i) Carry out the step for U = mgh with h = x·sinθ on a frictionless incline. Show that d/dt(½mv² + mgx·sinθ) = 0 reduces to the familiar a = −g·sinθ.

ii) For the roller coaster of item 3, the mechanical energy lost between 1 and 5 was 1960 J. If the friction force on the final climb is constant and the climb is 9.0 m of track, what is f? Why is the answer independent of how fast the marble was moving?

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